

## PROJECT 2: E-commerce Sales Analysis (Problems + Code ‘Python’)

```
[1]: import numpy as np
import pandas as pd
import mysql.connector
from sqlalchemy import create_engine

[2]: engine = create_engine('mysql+mysqlconnector://root:2000@localhost/ecommerce_sales')

[3]: query = 'select * from ecommerce_sales'

[16]: query = "SELECT * FROM indian_ecommerce"

df = pd.read_sql(query, engine)
df
```

|  | order_id | order_date | state     | zone           | category | brand_type         | customer_gender | customer_age | base_price | discount_percent | final_price | units_sold | r  |    |
|--|----------|------------|-----------|----------------|----------|--------------------|-----------------|--------------|------------|------------------|-------------|------------|----|----|
|  | 0        | ORD0000001 | 1/31/2023 | West Bengal    | East     | Fashion            | Mass            | Male         | 28         | 1810.89          | 65.00       | 633.81     | 29 | 1  |
|  | 1        | ORD0000002 | 1/31/2023 | Gujarat        | West     | Sports & Fitness   | Mass            | Male         | 19         | 5678.15          | 50.83       | 2791.95    | 67 | 18 |
|  | 2        | ORD0000003 | 1/31/2023 | Delhi NCR      | North    | Grocery Essentials | Mass            | Male         | 25         | 169.98           | 35.26       | 110.05     | 57 |    |
|  | 3        | ORD0000004 | 1/31/2023 | Madhya Pradesh | Central  | Footwear           | Mass            | Female       | 30         | 3244.91          | 61.94       | 1235.01    | 55 | 6  |
|  | 4        | ORD0000005 | 1/31/2023 | Haryana        | North    | Fashion            | Premium         | Female       | 33         | 697.44           | 48.27       | 360.79     | 29 | 1  |

|       |            |           |             |       |                  |         |        |    |          |       |         |    |    |
|-------|------------|-----------|-------------|-------|------------------|---------|--------|----|----------|-------|---------|----|----|
| 27921 | ORD0027922 | 9/30/2025 | Kerala      | South | Sports & Fitness | Mass    | Male   | 34 | 4717.50  | 63.56 | 1719.06 | 69 | 11 |
| 27922 | ORD0027923 | 9/30/2025 | Haryana     | North | Fashion          | Premium | Male   | 32 | 1086.67  | 13.82 | 936.49  | 38 | 3  |
| 27923 | ORD0027924 | 9/30/2025 | Tamil Nadu  | South | Electronics      | Mass    | Female | 61 | 3650.13  | 18.12 | 2988.73 | 20 | 5  |
| 27924 | ORD0027925 | 9/30/2025 | Maharashtra | West  | Electronics      | Premium | Female | 23 | 10078.34 | 25.84 | 7474.10 | 22 | 16 |
| 27925 | ORD0027926 | 9/30/2025 | Delhi NCR   | North | Sports & Fitness | Mass    | Male   | 32 | 2106.19  | 59.39 | 855.32  | 56 | 4  |

27926 rows × 16 columns

```
< df.shape
(27926, 16)
```

```
# Q1. Which product category generates the highest revenue?

(
df.groupby('category')['revenue']
.sum()
.sort_values(ascending=False)
.apply(lambda x: f'₹{x/1000000:.2f} Cr')
)
```

```
category
Premium Lifestyle    ₹59.18 Cr
Electronics          ₹47.53 Cr
Sports & Fitness     ₹25.95 Cr
Home & Living        ₹21.39 Cr
Footwear             ₹18.07 Cr
Fashion              ₹14.36 Cr
Beauty & Personal Care ₹9.23 Cr
Grocery Essentials   ₹2.81 Cr
Name: revenue, dtype: str
```

```
# Q2. Does high discount increase sales or reduce profit?
```

```
(
df.groupby('discount_percent')['revenue']
.mean()
.sort_index()
.apply(lambda x: f'₹{x/100000:.2f} Lakhs')
)
```

```
discount_percent
0.01      ₹0.34 Lakhs
0.02      ₹0.12 Lakhs
0.10      ₹0.08 Lakhs
0.14      ₹0.17 Lakhs
0.16      ₹0.17 Lakhs
...
64.96     ₹1.36 Lakhs
64.97     ₹0.73 Lakhs
64.98     ₹0.56 Lakhs
64.99     ₹0.81 Lakhs
65.00     ₹0.67 Lakhs
Name: revenue, Length: 5919, dtype: str
```

```
# Q3. Which states generate maximum sales revenue?
```

```
(
df.groupby('state')['revenue']
.sum()
.sort_values(ascending=False)
.apply(lambda x: f'₹{x/10000000:.2f} Cr')
)
```

```
state
Delhi NCR      ₹18.00 Cr
Maharashtra    ₹17.38 Cr
Karnataka      ₹17.26 Cr
Kerala         ₹15.49 Cr
Tamil Nadu     ₹15.06 Cr
Gujarat        ₹15.05 Cr
Telangana      ₹15.04 Cr
Punjab         ₹14.64 Cr
Haryana        ₹14.37 Cr
Madhya Pradesh ₹12.17 Cr
Uttar Pradesh  ₹11.23 Cr
West Bengal    ₹11.21 Cr
Odisha         ₹11.11 Cr
Chhattisgarh   ₹10.51 Cr
Name: revenue, dtype: str
```

```
# Q4. Does competition intensity affect revenue?
```

```
(
df.groupby('competition_intensity')['revenue']
.mean()
.apply(lambda x: f'₹{x/100000:.2f} Lakhs')
)
```

```
competition_intensity
High      ₹0.71 Lakhs
Low       ₹0.72 Lakhs
Medium    ₹0.70 Lakhs
Name: revenue, dtype: str
```

```
: # Q5. How does inventory pressure affect sales?
```

```
(  
df.groupby('inventory_pressure')['revenue']  
.mean()  
.apply(lambda x: f'₹{x/100000:.2f} Lakhs')  
)
```

```
: inventory_pressure  
High    ₹0.70 Lakhs  
Low     ₹0.72 Lakhs  
Name: revenue, dtype: str
```